

Операционные системы

Командная оболочка Midnight Commander

Сарханов Рамиэль Маркс оглы

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

2. Процесс выполнения лабораторной работы

Работа с Midnight Commander

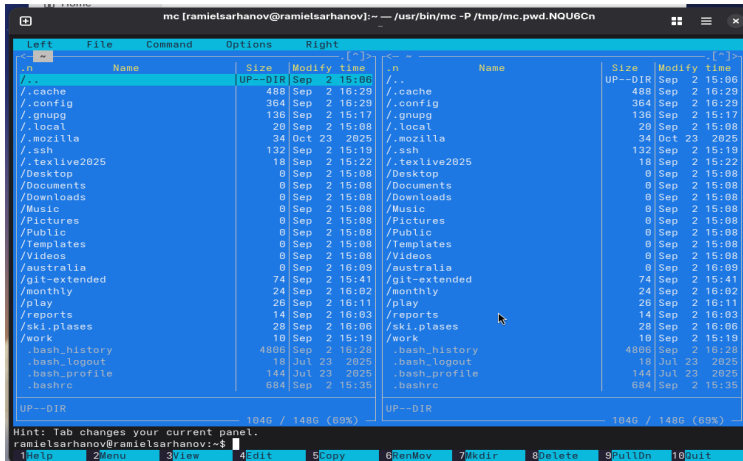


Рисунок 1: Запуск ms

Работа с Midnight Commander

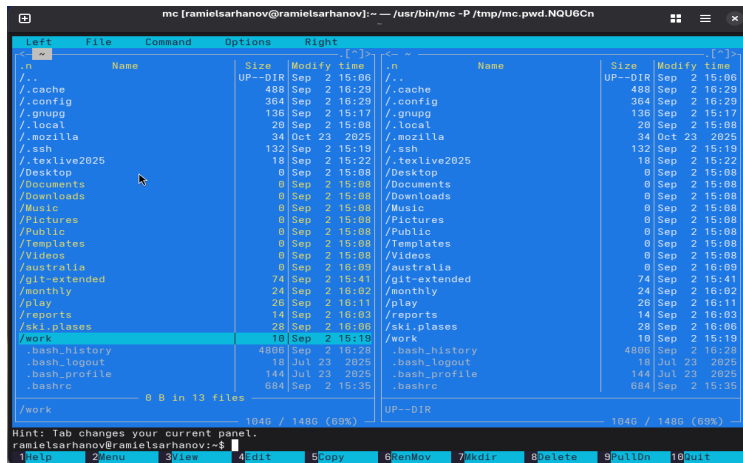


Рисунок 2: Выделение

Работа с Midnight Commander

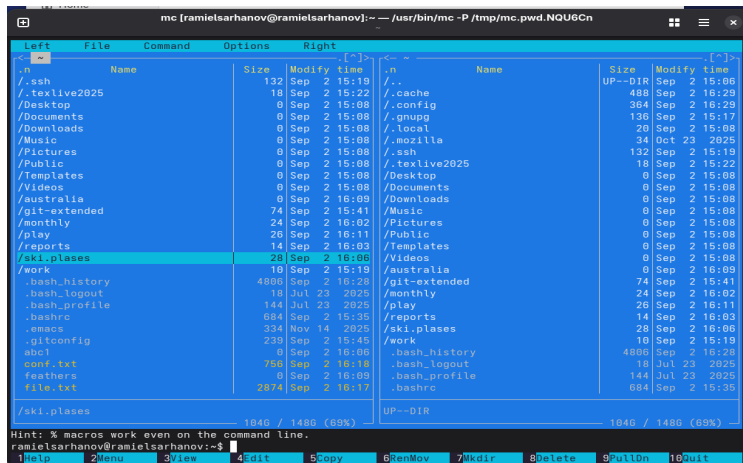


Рисунок 3: Отмена

Работа с Midnight Commander

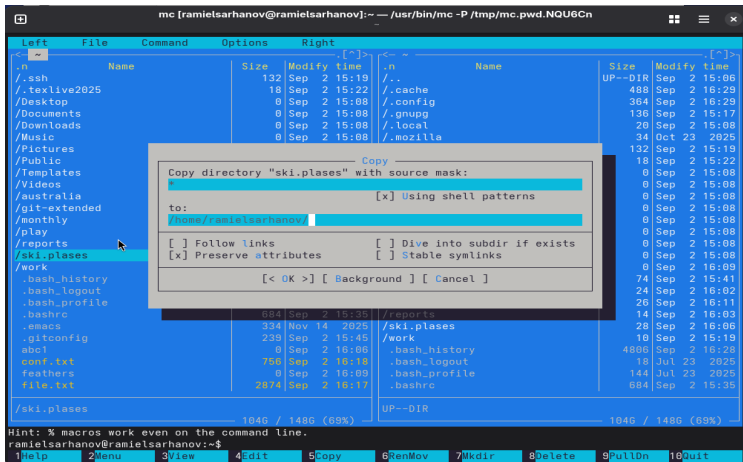


Рисунок 4: Копирование

Работа с Midnight Commander

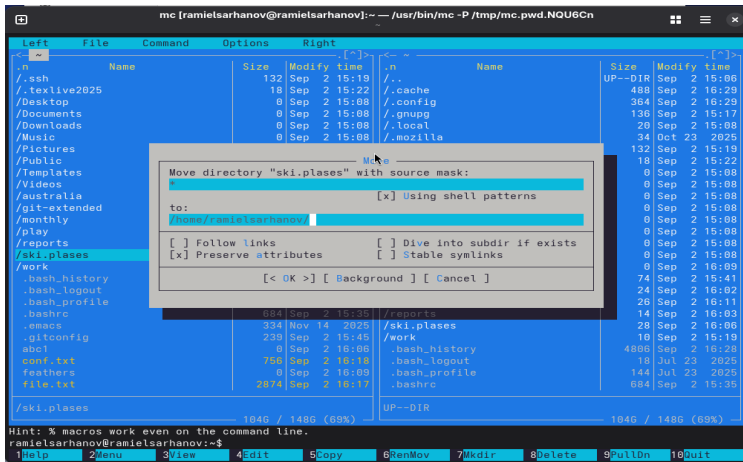


Рисунок 5: Перемещение

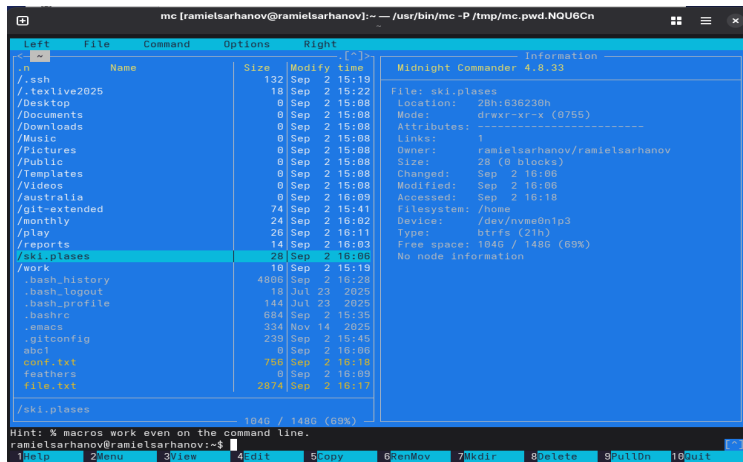


Рисунок 6: Информация

Работа с Midnight Commander

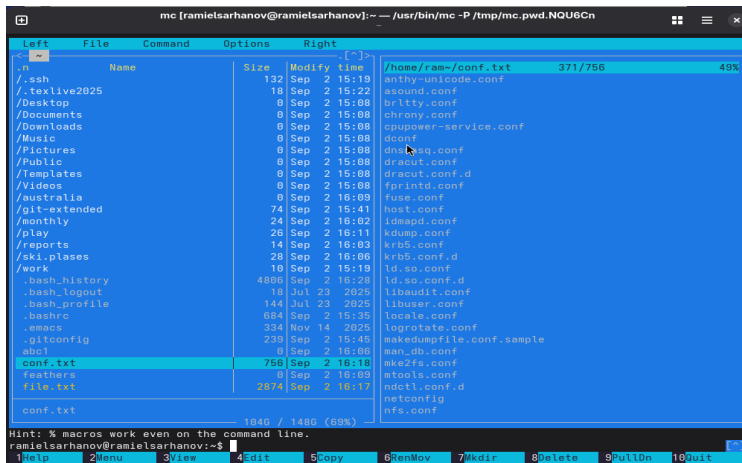


Рисунок 7: Быстрый просмотр

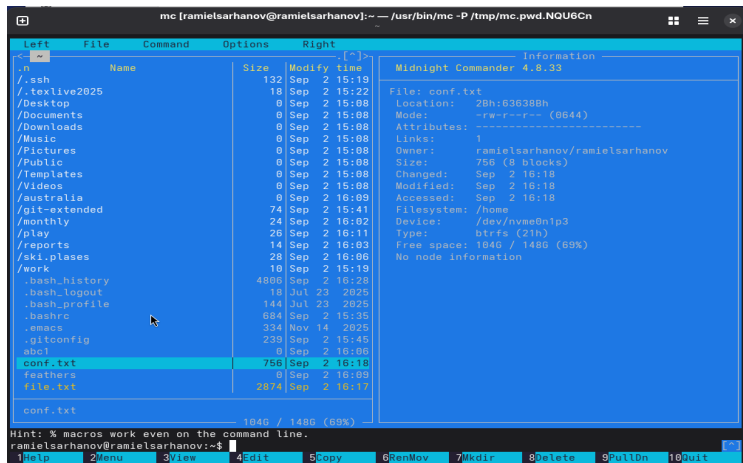


Рисунок 8: Информация

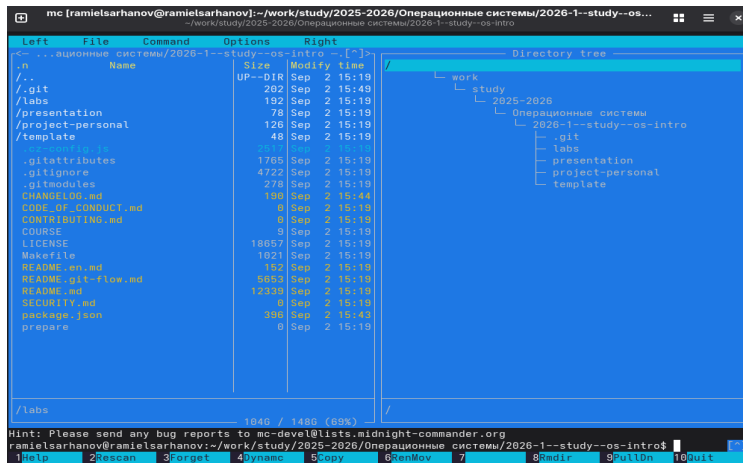
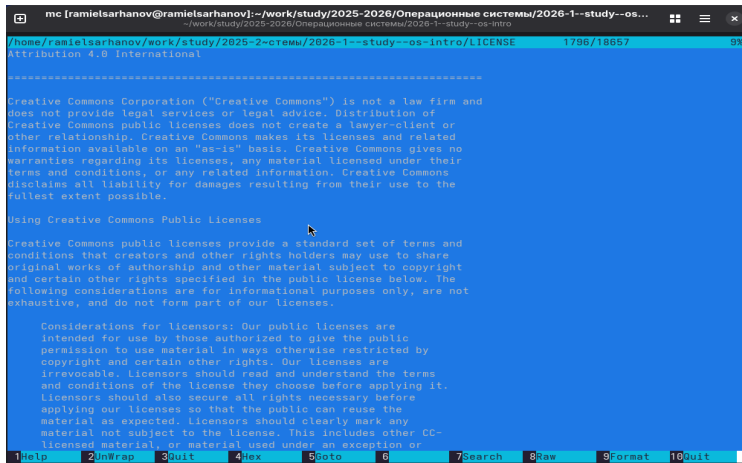


Рисунок 9: Дерево каталогов



The screenshot shows the Midnight Commander (MC) interface. The title bar indicates the user is 'ramielsarhanov' and the current directory is '/work/study/2025-2026/Операционные системы/2026-1--study--os...'. The main window displays the contents of a file named 'LICENSE' in the directory '/home/ramielsarhanov/work/study/2025-2026-Операционные системы/2026-1--study--os-intro/'. The file size is 1796/18657 bytes. The license text is displayed in a monospaced font on a blue background. At the bottom, there is a status bar with various shortcuts: 1Help, 2UnWrap, 3Quit, 4Hex, 5Goto, 6, 7Search, 8Raw, 9Format, 10Quit.

```
mc [ramielsarhanov@ramielsarhanov]:~/work/study/2025-2026/Операционные системы/2026-1--study--os...
~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/
/home/ramielsarhanov/work/study/2025-2026-Операционные системы/2026-1--study--os-intro/LICENSE 1796/18657 9%
Attribution 4.8 International

*****

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```

Рисунок 10: Просмотр содержимого текстового файла

Работа с Midnight Commander

```
mc [ramielsarhanov@ramielsarhanov]: ~/work/study/2025-2026/Операционные системы/2026-1--study--os...
--/work/study/2025-2026/Операционные системы/2026-1--study--os-intro

Makefile [----] 37 L: [ 1+18 19/ 28] *(637 /1021b) 0105 0x069 [*][X]
SHELL := /bin/bash
COURSE =.

.PHONY: all clean

all: help

help:
----->@echo 'Usage:'
----->@echo '  make <target>'
----->@echo.
----->@echo 'Targets:'
----->@grep -E '^ [a-zA-Z_0-9.-]+.*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BE
----->@grep -E '^###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $$1, $$
----->@grep -E '^ [a-zA-Z_0-9.-]+.*?###.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = ".*?
----->@echo

list:<-->## List of courses
----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
----->@. ./template/config/script/prepare
----->@touch prepare

submodule:<----->## Update submules
----->git submodule update --init --recursive
----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset -

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 11: Отредактируем содержимое текстового файла без сохранения результатов

Работа с Midnight Commander

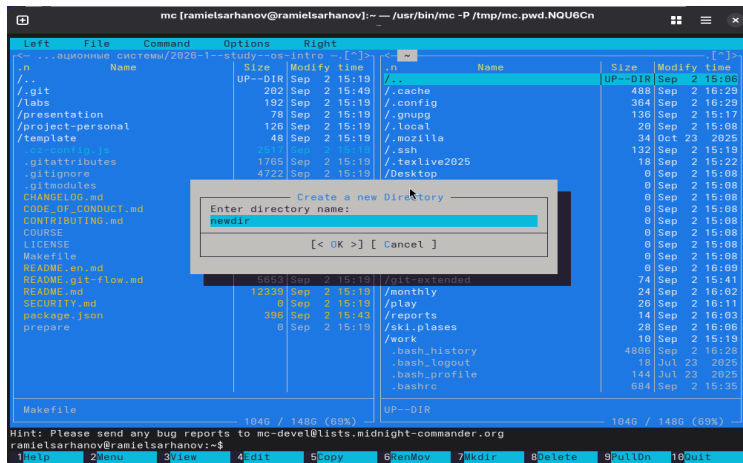


Рисунок 12: Создание каталога

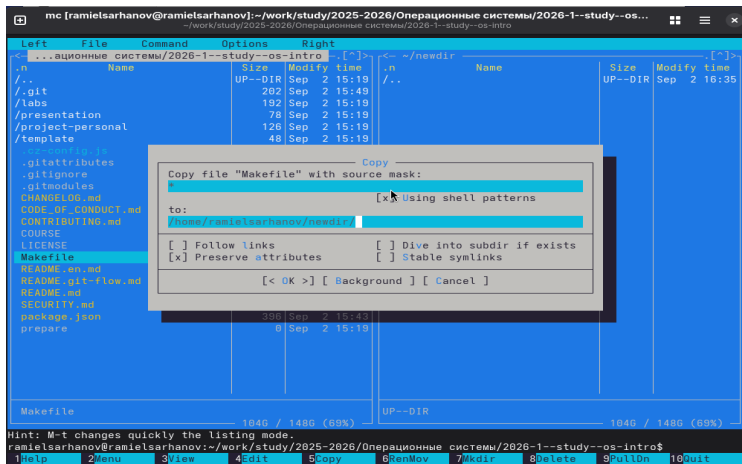


Рисунок 13: Копирование в файлов в созданный каталог

Работа с Midnight Commander

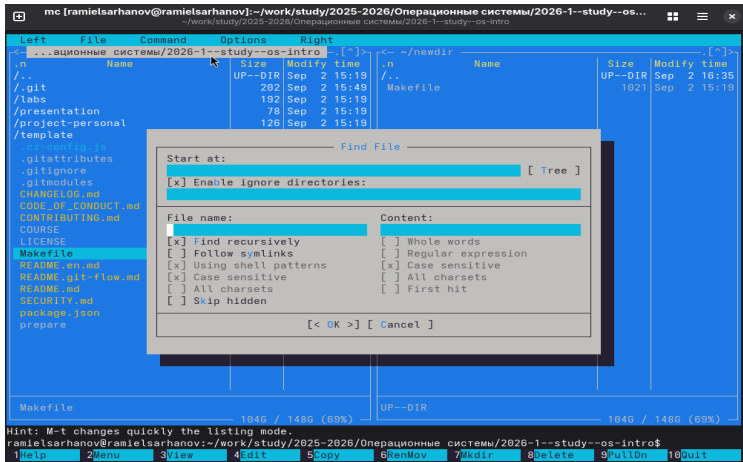


Рисунок 14: Поиск файлов

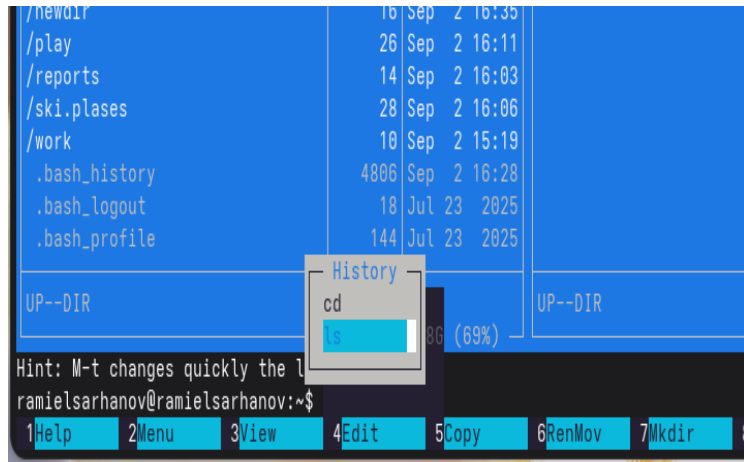


Рисунок 15: История команд

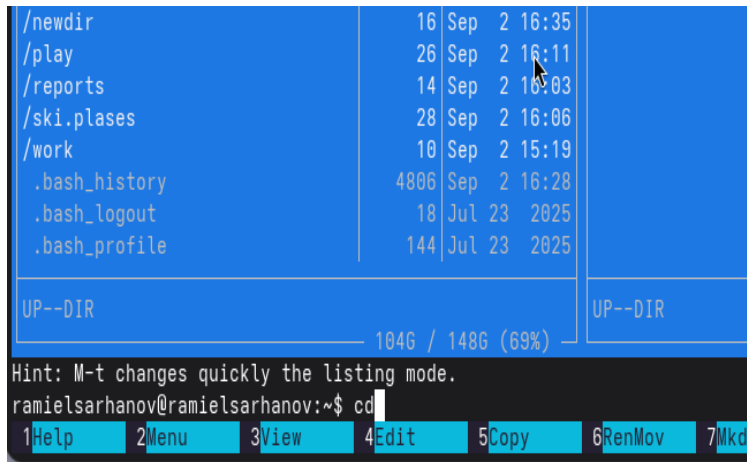
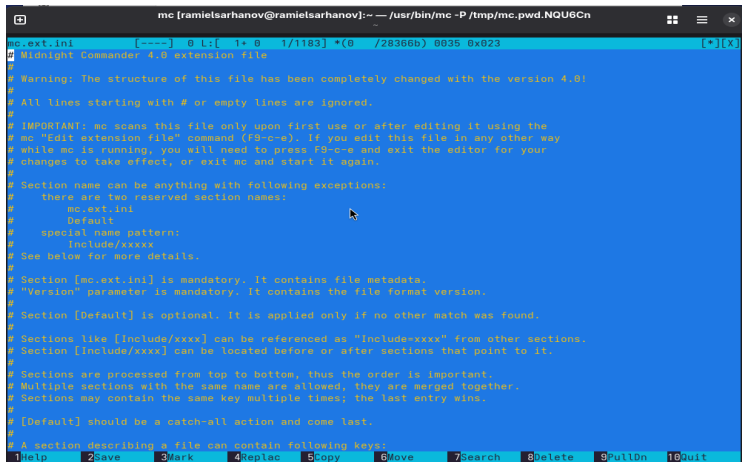


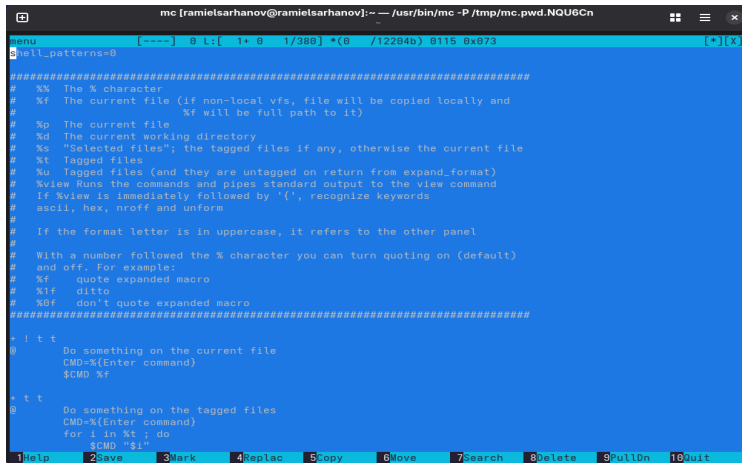
Рисунок 16: Переход в домашний каталог



The screenshot shows a terminal window titled "mc [ramielsarhanov@ramielsarhanov]: ~ — /usr/bin/mc -P /tmp/mc.pwd.NQU6Cn". The window displays the contents of the "mc.ext.ini" file, which is a configuration file for Midnight Commander. The file content includes a warning about structural changes in version 4.0, instructions on how to edit the file, and a list of reserved section names. At the bottom of the window, there is a status bar with various keyboard shortcuts: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, and 10Quit.

```
mc.ext.ini  [----]  0 L:[ 1+ 0 1/1183] *(0 /28366b) 0035 0x023  [*][X]
# Midnight Commander 4.0 extension file
#
# Warning: The structure of this file has been completely changed with the version 4.0!
#
# All lines starting with # or empty lines are ignored.
#
# IMPORTANT: mc scans this file only upon first use or after editing it using the
# mc "Edit extension file" command (F9-c-e). If you edit this file in any other way
# while mc is running, you will need to press F9-c-e and exit the editor for your
# changes to take effect, or exit mc and start it again.
#
# Section name can be anything with following exceptions:
#   there are two reserved section names:
#       mc.ext.ini
#       Default
#   special name pattern:
#       [include/xxxx]
# See below for more details.
#
# Section [mc.ext.ini] is mandatory. It contains file metadata.
# "Version" parameter is mandatory. It contains the file format version.
#
# Section [Default] is optional. It is applied only if no other match was found.
#
# Sections like [include/xxxx] can be referenced as "include=xxxx" from other sections.
# Section [include/xxxx] can be located before or after sections that point to it.
#
# Sections are processed from top to bottom, thus the order is important.
# Multiple sections with the same name are allowed, they are merged together.
# Sections may contain the same key multiple times; the last entry wins.
#
# [Default] should be a catch-all action and come last.
#
# A section describing a file can contain following keys:
1Help  2Save  3Mark  4Replac  5Copy  6Move  7Search  8Delete  9PullDn  10Quit
```

Рисунок 17: Просмотр файла расширений

The image shows a terminal window with the Midnight Commander interface. The title bar reads 'mc [ramielsarhanov@ramielsarhanov]:~ — /usr/bin/mc -P /tmp/mc.pwd.NQU6Cn'. The status bar at the top shows 'enu [----] 0 L: [1+ 0 1/380] *(0 /12204b) 0115 0x073 [*][X]'. The main area displays the file menu, starting with 'shell_patterns=0'. It lists various macros and their functions, such as '%f' for the current file, '%p' for the current panel, and '%d' for the current directory. It also includes instructions on how to use the '!' and 't' characters to execute commands on the current file or tagged files. At the bottom, a menu bar lists numbered options: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, and 10Quit.

```
mc [ramielsarhanov@ramielsarhanov]:~ — /usr/bin/mc -P /tmp/mc.pwd.NQU6Cn
enu [----] 0 L: [ 1+ 0 1/380] *(0 /12204b) 0115 0x073 [*][X]
shell_patterns=0

#####
# %% The % character
# %f The current file (if non-local vfs, file will be copied locally and
#      %f will be full path to it)
# %p The current file
# %d The current working directory
# %s "Selected files"; the tagged files if any, otherwise the current file
# %t Tagged files
# %u Tagged files (and they are untagged on return from expand_format)
# %view Runs the commands and pipes standard output to the view command
# If %view is immediately followed by '(', recognize keywords
#      ascii, hex, nroff and unform
#
# If the format letter is in uppercase, it refers to the other panel
#
# With a number followed the % character you can turn quoting on (default)
# and off. For example:
# %f   quote expanded macro
# %!f  ditto
# %0f  don't quote expanded macro
#####

+ ! t t
@      Do something on the current file
      CMD=%{Enter command}
      $CMD %f

+ t t
@      Do something on the tagged files
      CMD=%{Enter command}
      for i in %t ; do
      $CMD "$i"

1Help    2Save    3Mark    4Replac  5Copy    6Move    7Search  8Delete  9PullDn 10Quit
```

Рисунок 18: Просмотр файла меню

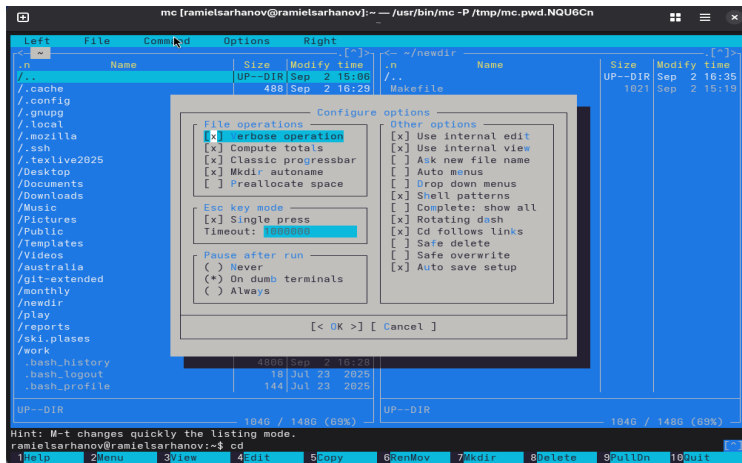


Рисунок 19: Конфигурация

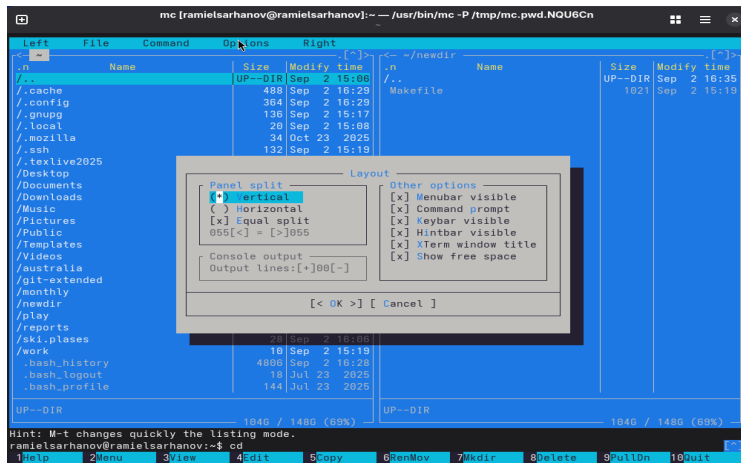


Рисунок 20: Внешний вид

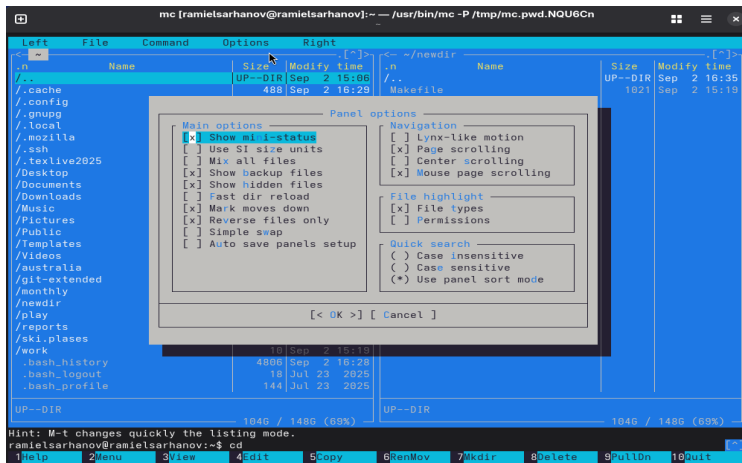


Рисунок 21: Настройки панелей

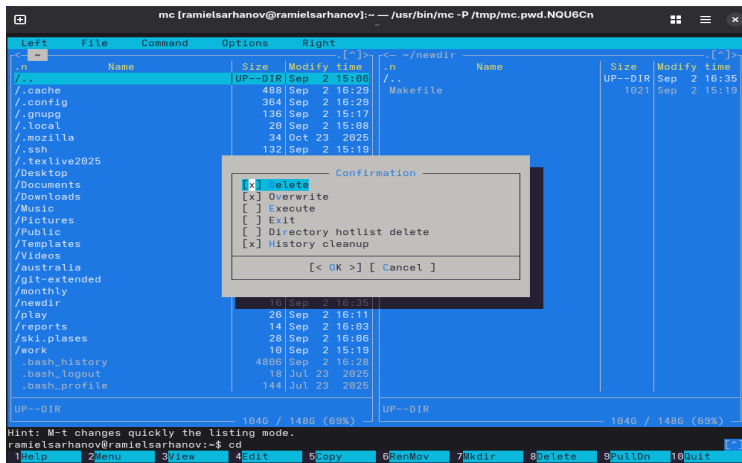


Рисунок 22: Подтверждение

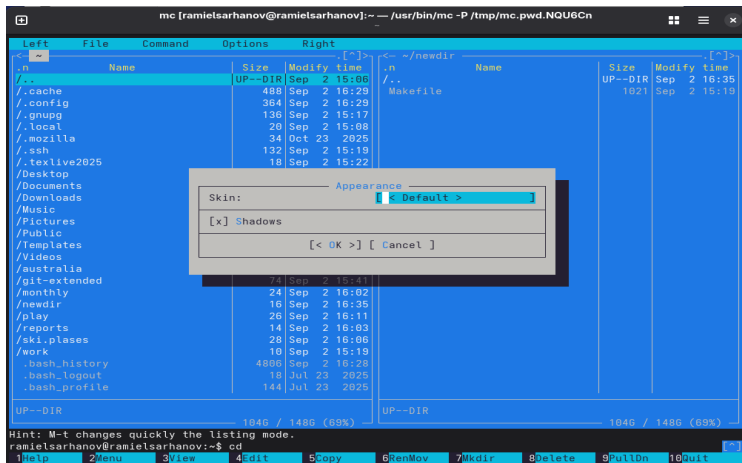


Рисунок 23: Оформление

Работа с Midnight Commander

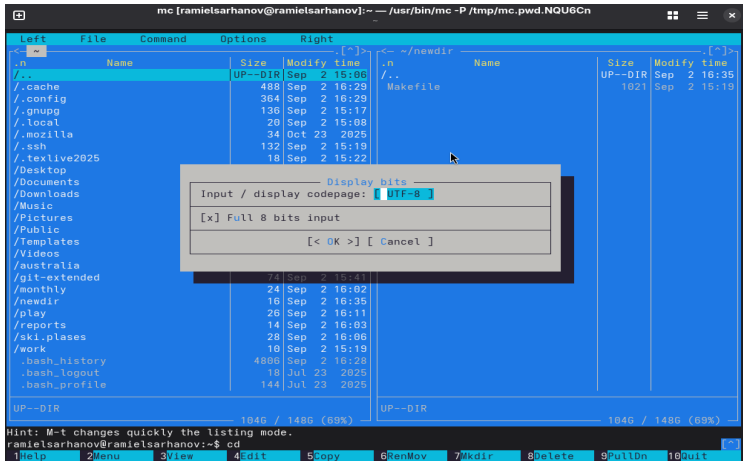


Рисунок 24: Кодировка символов

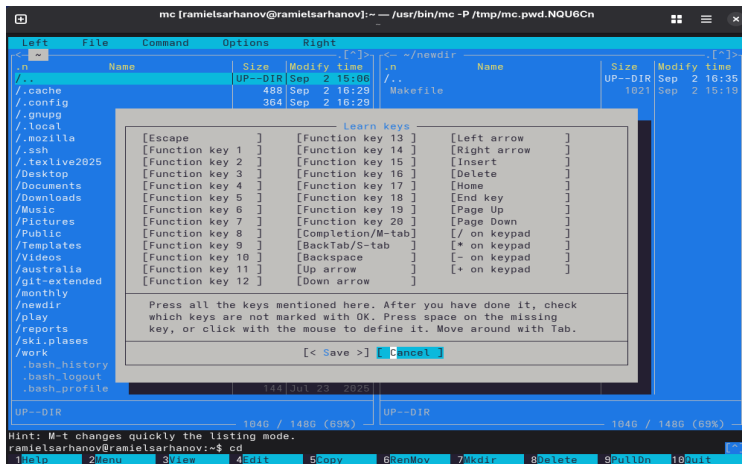
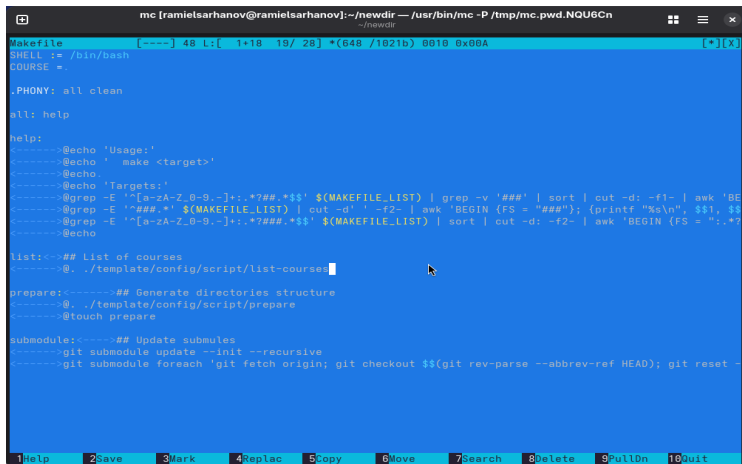


Рисунок 25: Распознавание клавиш

Работа с редактором Midnight Commander



```
mc [ramielsarhanov@ramielsarhanov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.NQU6Cn
~/newdir
Makefile  [----] 48 L: [ 1+18 19/ 28] *(648 /1021b) 0010 0x00A [*][X]
SHELL := /bin/bash
COURSE =.

.PHONY: all clean

all: help

help:
----->@echo 'Usage:'
----->@echo '  make <target>'
----->@echo.
----->@echo 'Targets:'
----->@grep -E '^[a-zA-Z_0-9.-]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BE
----->@grep -E '###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $$1, $$
----->@grep -E '^[a-zA-Z_0-9.-]+:.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = ".*?
----->@echo

list:<-->## List of courses
----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
----->@. ./template/config/script/prepare
----->@touch prepare

submodule:<----->## Update submules
----->git submodule update --init --recursive
----->git submodule foreach 'git fetch origin; git checkout $$$(git rev-parse --abbrev-ref HEAD); git reset -

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 26: Файл с текстом

Работа с редактором Midnight Commander

```
mc [ramielsarhanov@ramielsarhanov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.NQU6Cn
~/newdir
Makefile [-M--] 0 L: [ 1+15 16/ 25] *(574 / 988b) 0009 0x009 [*](x)
SHELL := /bin/bash
COURSE =.

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo.
<----->@echo 'Targets:'
<----->@grep -E '^[a-zA-Z_0-9.-]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BE
<----->@grep -E '###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $$1, $$
<----->@grep -E '^[a-zA-Z_0-9.-]+:.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = ":.??
<----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare

submodule:<----->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset --
```

Рисунок 27: Файл с текстом

Работа с редактором Midnight Commander

```
mc [ramielsarhanov@ramielsarhanov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.NQU6Cn
~/newdir
Makefile [BM--] 0 L: [ 1+18 19/ 34] *(660 /1246b) 0009 0x009 [*][X]
SHELL := /bin/bash
COURSE = .
.PHONY: all clean
all: help
help:
----->@echo 'Usage:'
----->@echo ' make <target>'
----->@echo
----->@echo 'Targets:'
----->@grep -E '^[a-zA-Z_0-9.-]+.*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BE
----->@grep -E '^[a-zA-Z_0-9.-]+.*?###.*$$' $(MAKEFILE_LIST) | cut -d: -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $$1, $$
----->@grep -E '^[a-zA-Z_0-9.-]+.*?###.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $$1, $$
----->@. ./template/config/script/list-courses
prepare:----->## Generate directories structure
----->@. ./template/config/script/list-courses
prepare:----->## Generate directories structure
----->@. ./template/config/script/list-courses
prepare:----->## Generate directories structure
----->@. ./template/config/script/prepare
----->@touch prepare
submodule:----->## Update submules
----->git submodule update --init --recursive
----->git submodule foreach 'git fetch origin; git checkout $(git rev-parse --abbrev-ref HEAD); git reset --
1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 28: Копирование фрагмента

Работа с редактором Midnight Commander

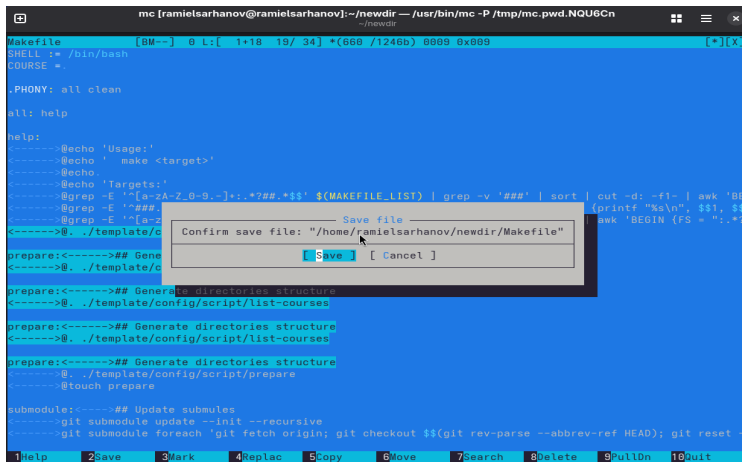
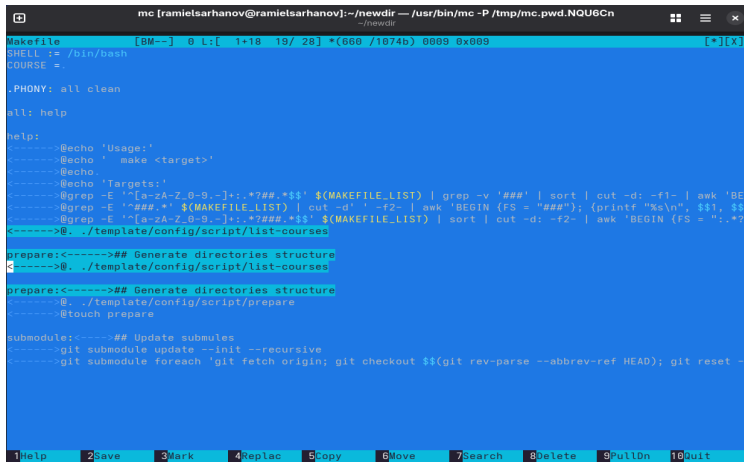


Рисунок 29: Сохранение

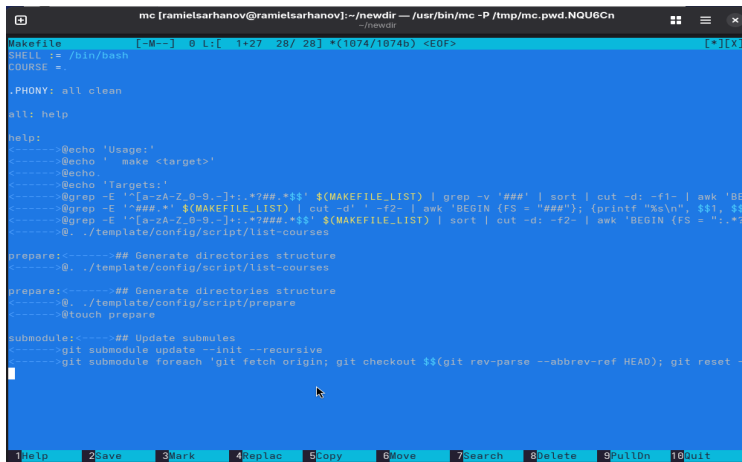
Работа с редактором Midnight Commander



```
mc [ramietsarhanov@ramietsarhanov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.NQU6Cn
~/newdir
Makefile [BM--] 0 L: [ 1+18 19/ 28] *(660 /1074b) 0009 0x009 [*][X]
SHELL := /bin/bash
COURSE =.
.PHONY: all clean
all: help
help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo.
<----->@echo 'Targets:'
<----->@grep -E '^ [a-zA-Z_0-9.-]+.*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BE
<----->@grep -E '^###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $$1, $$
<----->@grep -E '^ [a-zA-Z_0-9.-]+.*?###.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = ".*?
<----->@. ./template/config/script/list-courses
prepare:<----->## Generate directories structure
<----->@. ./template/config/script/list-courses
prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare
submodule:<----->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset -
```

Рисунок 30: Отмена

Работа с редактором Midnight Commander



```
mc [ramielsarhanov@ramielsarhanov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.NQU6Cn
~/newdir
Makefile  [-M--]  0 L:[ 1+27 28/ 28] *(1074/1074b) <EOF> [*][X]
SHELL := /bin/bash
COURSE =.

.PHONY: all clean

all: help

help:
----->@echo 'Usage:'
----->@echo '  make <target>'
----->@echo.
----->@echo 'Targets:'
----->@grep -E '^[a-zA-Z_0-9.-]+.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BE
----->@grep -E '###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $$1, $$
----->@grep -E '^[a-zA-Z_0-9.-]+.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = ".*?
----->@. ./template/config/script/list-courses

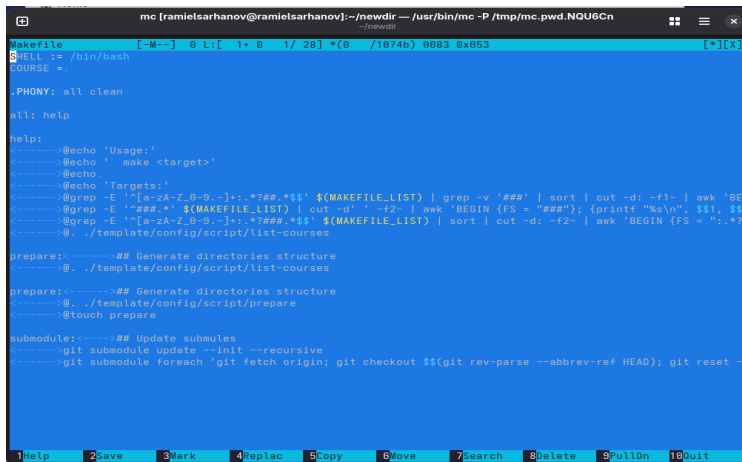
prepare:----->## Generate directories structure
----->@. ./template/config/script/list-courses

prepare:----->## Generate directories structure
----->@. ./template/config/script/prepare
----->@touch prepare

submodule:----->## Update submodules
----->git submodule update --init --recursive
----->git submodule foreach 'git fetch origin; git checkout $$$(git rev-parse --abbrev-ref HEAD); git reset -
```

Рисунок 31: Переход в конец файла

Работа с редактором Midnight Commander



```
mc [ramielsarhanov@ramielsarhanov]:~/newdir — /usr/bin/mc -P tmp/mc.pwd.NQU6Cn
~/newdir
Makefile      [ -M-- ]  0 L:[ 1+ 0 1/ 20 ] *(0 /1074b) 0083 0x053 [*](X)
SHELL := /bin/bash
COURSE = .
.PHONY: all clean
all: help
help:
----->@echo 'Usage:'
----->@echo '  make <target>'
----->@echo.
----->@echo 'Targets:'
----->@grep -E '^[a-zA-Z_0-9.-]+.*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BE
----->@grep -E '###.*' $(MAKEFILE_LIST) | cut -d: -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $$1, $$
----->@grep -E '^[a-zA-Z_0-9.-]+.*?###.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = ".*?
----->@. ./template/config/script/list-courses

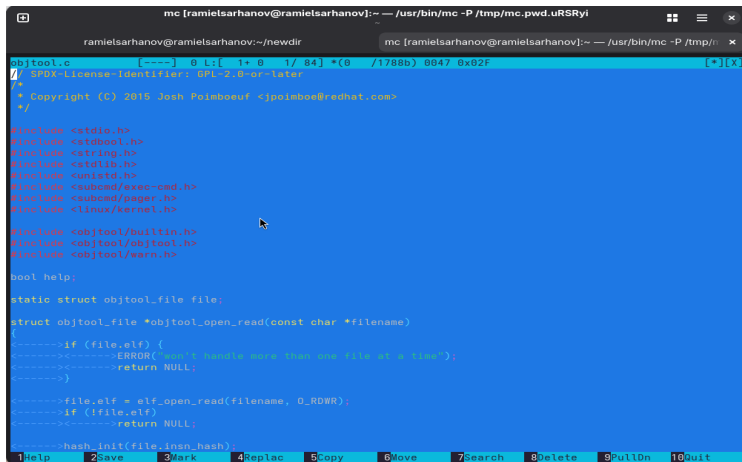
prepare:----->## Generate directories structure
----->@. ./template/config/script/list-courses

prepare:----->## Generate directories structure
----->@. ./template/config/script/prepare
----->@touch prepare

submodule:----->## Update submodules
----->git submodule update --init --recursive
----->git submodule foreach 'git fetch origin; git checkout $$$(git rev-parse --abbrev-ref HEAD); git reset --
```

Рисунок 32: Переход в начало файла

Работа с редактором Midnight Commander



```
mc [ramielsarhanov@ramielsarhanov]:~ — /usr/bin/mc -P /tmp/mc.pwd.uRSRyi
ramielsarhanov@ramielsarhanov:~/newdir mc [ramielsarhanov@ramielsarhanov]:~ — /usr/bin/mc -P /tmp/mc.pwd.uRSRyi
objtool.c [----] 0 L: [ 1+ 0 1/ 84] *(0 /1788b) 0047 0x02F [*][X]
/* SPDX-License-Identifier: GPL-2.0-or-later */
/* Copyright (C) 2015 Josh Poimboeuf <jpoimboe@redhat.com> */

#include <stdio.h>
#include <stdbool.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>
#include <subcmd/exec-cmd.h>
#include <subcmd/pager.h>
#include <linux/kernel.h>

#include <objtool/builtin.h>
#include <objtool/objtool.h>
#include <objtool/warn.h>

bool help;

static struct objtool_file file;

struct objtool_file *objtool_open_read(const char *filename)
{
    -----> if (file.elf) {
    ----->-----> ERROR("can't handle more than one file at a time");
    ----->-----> return NULL;
    -----> }

    -----> file.elf = elf_open_read(filename, O_RDWR);
    -----> if (!file.elf)
    ----->-----> return NULL;

    -----> hash_init(file.insn_hash);
1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 33: Файл с программой

Работа с редактором Midnight Commander

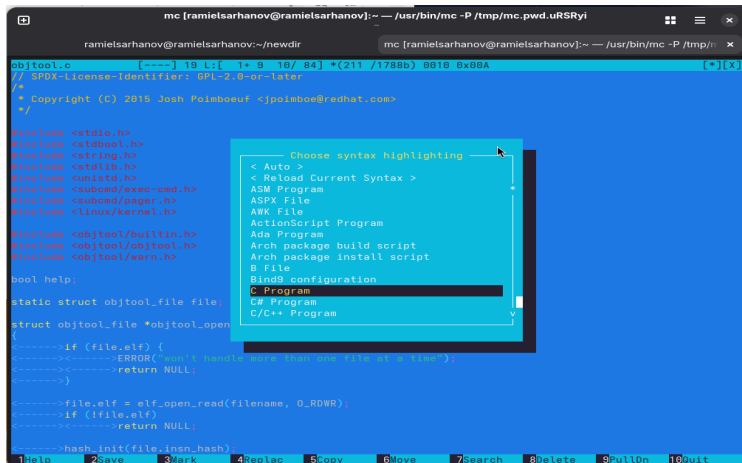


Рисунок 34: Цветовыделение синтаксиса

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.